Preview

Score for this attempt: **4** out of 4

Question 1

1 / 1 pts

You have a deck of 52 cards and you deal 5 cards (one-by-one). Find the conditional probability that the last card is a king given that the other cards are queens.

Provide a numerical answer with precision up to 4 decimal places.

0.0833

After dealing 4 queens (in particular 4 cards that are not kings), there are 48 cards left, of which 4 are kings.

$$P(\text{king} \mid \text{saw 4 queens}) = \frac{4}{48} = \frac{1}{12} \approx 0.0833$$

Question 2

1 / 1 pts

Suppose $P(A) = 0.27$, $P(B) = 0.38$, and $P(A|B) = 0.17$. Compute $P(B|A)$.

Provide a numerical solution with precision up to 3 decimal places.

0.239

By Bayes' rule,

$$P(B|A) = \frac{P(A|B) \cdot P(B)}{P(A)} = \frac{0.17 \times 0.38}{0.27}.$$

Question 3

1 / 1 pts

We choose a number uniformly at random from the set $\{1, 2, \ldots, 18\}$. Consider the following events:

A={the chosen number is divisible by 2} $= \{2, 4, 6, 8, 10, 12, 14, 16, 18\} \longrightarrow P(A) = \frac{9}{18} = \frac{1}{2}$

B={the chosen number is divisible by 3} $= \{3, 6, 9, 12, 15, 18\} \longrightarrow P(B) = \frac{6}{18} = \frac{1}{3}$

C={the chosen number is divisible by 7} $= \{7, 14\} \rightarrow P(C) = \frac{2}{18} = \frac{1}{9}$

Which of the following statements are true?

Note: $A \cap B = \{6, 12, 18\}$
$A \cap C = \{14\}$
$B \cap C = \emptyset$

- [ ] A, B, and C are independent — False because $P(A \cap B \cap C) = 0$ but none of $P(A)$, $P(B)$, or $P(C)$ is 0.
- [x] A and B are independent — True because $P(A \cap B) = \frac{3}{18} = \frac{1}{6} = \frac{1}{2} \times \frac{1}{3} = P(A)P(B)$.
- [ ] B and C are independent — False because $B \cap C = \emptyset \rightarrow P(B \cap C) = 0$. However, $P(B)$ and $P(C)$ are nonzero.
- [x] A and C are independent — True because $P(A \cap C) = \frac{1}{18} = \frac{1}{2} \times \frac{1}{9} = P(A)P(C)$.

Question 4

1 / 1 pts

Three balls are selected, one at a time and without replacement, from a urn that contains 1 red ball, 1 yellow ball, and 1 green ball. Consider three random variables:

- X is the random variable that returns 1 if the green or red ball is selected first and returns 0 otherwise,
- Y is the random variable that returns 1 if the yellow ball is selected first and returns 0 otherwise;
- Z is the random variable that returns 1 if the red ball is selected before the green ball is selected and returns 0 otherwise.

Which of the following statements are true?

a. ☑ X and Z are independent

b. ☐ X, Y, Z are independent — False because $P(X=1, Y=1, Z=1) = 0$ but none of $P(X=1)$, $P(Y=1)$, or $P(Z=1)$ is zero.

c. ☐ X and Y are independent — False because $P(X=1 \text{ and } Y=1) = 0$ but $P(X=1) \neq 0$ and $P(Y=1) \neq 0$

d. ☑ Y and Z are independent

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To verify that statements a and d are true, we model the problem as follows: set

$$\Omega = \left\{ \begin{matrix} RGY, & RYG, \\ GRY, & GYR, \\ YRG, & YGR \end{matrix} \right\}.$$

(Alternatively, we could number the balls.)

Our space Ω consists of six outcomes, each equally likely.

Note that $\{X=1\} = \{RGY, RYG, GRY, GYR\}$ $\quad P(X=1) = \frac{4}{6} = \frac{2}{3}$

$\{Y=1\} = \{YRG, YGR\}$ $\quad P(Y=1) = \frac{1}{3}$

$\{Z=1\} = \{RGY, RYG, YRG\}$ $\quad P(Z=1) = \frac{1}{2}$

Define $A := \{X=1\}$, $B := \{Y=1\}$, $C := \{Z=1\}$. The events A and C are independent, so X and Z are independent, i.e. statement a is true. Similarly, the events B and C are independent so statement d is true.